

Naoufal Acharki

PHD IN STATISTICS AND MACHINE LEARNING · SENIOR DATA SCIENTIST

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Summary

PhD in Applied Mathematics with 7+ years of experience in Data Science and Machine Learning. Specialized in designing, building, and deploying end-to-end ML solutions for recommendation systems, from PoC to production, with a strong focus on experimentation and causal inference.

Professional Experience

Air France-KLM – Commercial Data Officer - Major airline group, European air transport leader

Roissy-en-France

SENIOR DATA SCIENTIST - MARKETING OPERATIONS RESEARCH

Nov 2025 - Present

- Contributed to 'Offer Display Recommender' project to personalize the customer product offering
- Designed and executed A/B tests to optimize the customer journey and experience on the website, particularly for flight selection
- Analyzed customer behavior and translated insights into strategies/decisions for business stakeholders.

Mercor – Recruitment platform connecting talent with AI labs

San Francisco, USA

MATHEMATICS/STATISTICS EXPERT

Aug 2025 - Jul 2026

- Wrote mathematical problems to challenge the reasoning capabilities of state-of-the-art language models (LLMs).

Senzai – Data infrastructure and analytics startup, optimizing marketing campaigns

Mexico City, Mexico (Remote)

SENIOR DATA SCIENTIST AND MACHINE LEARNING ENGINEER - PIPELINES AND MLOPS LEAD

Oct 2023 - Oct 2025

- Designed a recommendation system to optimize customer targeting during digital marketing campaigns, **increasing conversion rates by up to 30%** across **3 million customers** and **50 million interactions**.
- Designed and deployed data engineering modules and causal inference ML methods/models into production.
- Established **MLOps practices and scalable backend pipelines** (GitLab CI/CD, Docker on AWS EC2, Kubernetes, Airflow, MLflow)

namR – Data intelligence startup specializing in buildings and the energy transition.

Paris, France

DATA SCIENTIST - GEO & NLP TEAM

May 2023 - Oct 2023

- Developed new features on large-scale datasets (up to 50 million rows on GCP).
- Optimized existing ML algorithms, reducing solar panel detection runtime **from 25 to 8 hours**.

TotalEnergies One Tech – R&D branch of the French energy giant TotalEnergies

Palaiseau, France

RESEARCH ENGINEER - AI & DATA SCIENCE R&D PROGRAM

Oct 2019 - Dec 2022

- Developed Gaussian process regression models to predict gas well production (80% confidence) and causal inference models for geothermal wells, **resulting in a patent filing and three scientific publications**.
- Presented research advances to Digital Factory stakeholders, delivering actionable insights and strengthening cross-team collaboration on data science initiatives.

Education

École Polytechnique, Institut Polytechnique de Paris

Palaiseau, France

PHD IN APPLIED MATHEMATICS – SPECIALIZATION: CAUSAL INFERENCE AND UNCERTAINTY QUANTIFICATION

Oct 2019 - Dec 2022

Skills: Statistics, Bayesian Modeling, Uncertainty Quantification, Causal Inference, Optimization, and Interpretability.

École des Mines de Saint-Étienne

Saint-Étienne, France

ENGINEERING DEGREE – MAJOR: DATA SCIENCE; MINOR: OPERATIONS RESEARCH

Sept 2016 - Sept 2019

Skills: Supervised and Unsupervised Learning, Time Series, High Performance Computing (HPC), Multi-Objective Optimization.

Université Paris-1 Panthéon Sorbonne

Paris, France

M.Sc. IN MATHEMATICAL MODELLING AND METHODS IN ECONOMICS AND FINANCE

Sept 2018 - Sept 2019

Skills: Stochastic Calculus, Arbitrage Theory, Optimization, Credit Risk, Financial Markets

Skills

Mathematics	Statistics, Machine Learning, Causal Inference, Optimization, Experimentation, Recommendation Systems
Programming	Python, R, SQL, MATLAB
Data Handling	Azure (TotalEnergies), GCP: BigQuery (namR, AFKLM), Vertex.ai (Senzai), AWS (Senzai, Kemba), PostgreSQL, MySQL, Trino
Tools	VSCode, PyCharm, Jupyter Notebook, Jira, Linear, Atlassian, Confluence, Kanban, Dataiku DSS, Grafana
Libraries	ggplot2, numpy, pandas, matplotlib, sqlalchemy, Scikit-learn, XGBoost, LightGBM, Shapely, CausalML, google-generativeai
MLOps	Docker, Airflow, DVC, MLflow, GitLab CI/CD, Streamlit, FastAPI, PySpark, AWS EC2 and Redshift